MINGKE (BRITNEY) TIAN

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EDUCATION

B.S. in Applied Mathematics and Statistics, B.S. in Computer Science **Aug. 2023 - May 2027**
Emory University; Emory College of Arts and Science *Atlanta, GA*

- ***Cumulative GPA****: 3.983/4.00*; Honor: Dean's List (2023-2026), Honor Thesis Program
- **Coursework**: Graph Theory (Graduate Level), Analysis Seminar (Grad Level), Real Analysis, Numerical PDEs, Numerical Analysis, Mathematical Statistics, Numerical Linear Algebra, Machine Learning, Mathematics of Deep Learning, Computer Vision, Data Structure & Algorithms, Computer Architecture, Operating Systems, HCI

PUBLICATIONS AND MANUSCRIPTS

[1] ***Mingke Tian***, Yiyun Chen, Ruipeng Li, Tianshi Xu, Yuanzhe Xi. *ATSolver: Agentic Tuning of Sparse Linear Solvers with Numerical Feedback*. ***International Conference on Learning Representations (ICLR 2027).***
First author | Under review [arxiv]

[2] ***Mingke Tian****, Riyang Bao*, et al. *MobilityWorld: Learning a Structured World Model for Human Mobility via Latent State-Action Dynamics*. ***AAAI Conference on Artificial Intelligence (AAAI 2027).***
Co-First author | Under review [arxiv]

[3] Riyang Bao*, ***Mingke Tian****, Andreas Zufle, Liang Zhao. *StayBERT: Staypoint Detection as the Pretext Task for Trajectory Foundation Models*. ***AAAI Conference on Artificial Intelligence (AAAI 2027).***
Co-First author | Under review [paper]

[4] Zeeshan Memon, ***Mingke Tian,*** Xinyuan Song, Hongwei Jin, Kibaek Kim, Liang Zhao. *CONSTRAINER: Promptable Graph-Structured Optimization via Constraint Conditioning.* ***Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026).***
Second author | Accepted to the main track, to appear. [paper]

RESEARCH EXPERIENCE

CONSTRAINER: Promptable Graph-Structured Optimization via Constraint Conditioning Sep. 2025 - Feb. 2026
Department of Computer Science & Argonne National Laboratory *Atlanta, GA*

- Proposed a constrained-conditioned foundation model for graph-structured optimization, enabling zero-shot transfer and generalization to power-grid optimization and planning without specific retraining.
- Designed an operator-DAG constraint encoder and implemented heterogeneous graph transformer cross-attention to preserve constraint semantics; reduced optimality gap by 45% and achieving 99.02% feasibility.
- Built scalable training along 16 combinatorial tasks; reached convergence in approx. 30% fewer epochs.

ATSolver: Agentic Tuning of Sparse Linear Solvers with Numerical Feedback **Jul. 2026 - Oct. 2026**
Department of Mathematics & Lawrence Livermore National Laboratory *Atlanta, GA*

- Developed a verifier-grounded agentic framework that uses execution feedback to improve sparse-solver decisions for entire workloads, reducing setup and solving costs while enforcing accuracy and resource limits.
- Designed a typed policy for LLM reasoning agents using workload cards, matrix views, memory, and retrieved knowledge and experience to coordinate solver–preconditioner selection, parameter tuning, and setup reuse.
- Built a gated propose–execute–diagnose–repair loop with independent residual checks and budget control.
- Raised solve success from 61% to 71.4% vs. equal-budget search, with 30.1% lower replayed solve time.

MobilityWorld: Learning a Structured World Model for Human Mobility via Latent State-Action Dynamics **Mar. 2026 - Jul. 2026**

- Developed a geographically grounded world model that learns human mobility as a decision-driven process from

sparse, action-unlabeled GPS, enabling inference, generation, and planning tasks through shared dynamics.
- Designed structured consistency learning over a probabilistic graph model to jointly recover latent state and action.
- Evaluated eight mobility tasks with profile-conditioned action priors, road–POI context, and graph-legal execution.

StayBERT: Staypoint Detection as the Pretext Task for Trajectory Foundation Models **Mar. 2026 - Jul. 2026**
- Developed a trajectory foundation model pretrained through supervised staypoint detection on raw GPS.
- Fused Transformer motion and place features via dimension-wise gate; decoded staypoint spans with CRF.
- Evaluated frozen features on seven tasks across three datasets; achieved 89.8% activity F1 in cross-city transfer.

Structured MAGI with Warm Start for Partially Observed Bayesian ODE Inverse Problem **Jun. 2025 - Dec. 2025**
Emory NSF REU Computational Mathematics for Data Science *Atlanta, GA*
- Developed physics-informed Bayesian inference for partially observed ODEs, jointly estimating latent trajectories and parameters from sparse and noisy observations with uncertainty quantification.
- Combined multi-task GP denoising with staged PINN estimates to initialize HMC and reduce burn-in steps.
- Analyzed Matérn kernel matrices for HSS structure and numerical rank to support Gaussian process inference.

WORK EXPERIENCE

Ricoh USA **Sep 2025 - Dec 2025**
AI/Machine Learning Engineer Intern, Co-op *Atlanta, GA*
- Built Qwen2.5-VL-7B extraction on SageMaker, fusing OCR/layout for ~600 enterprise documents (3.2K pages).
- LoRA-tuned labeled page groups, raising F1 from 86.7% to 92.3% and schema validity from 93.4% to 98.6%.
- Built an MLflow evaluation pipeline; prompt caching cut processing time 20.2% and F1 within 0.3 points of peak.

Siasun Robotics & Automation Co., Ltd. **Jun 2026 - Aug 2026**
Robotics ML Intern *China*
- Fine-tuned NVIDIA Isaac GR00T N1.7 on demos using PyTorch and deployed for real-robot parcel manipulation.
- Mapped normalized 16-step action chunks to relative EEF deltas and gripper commands for robot control.
- Applied DAgger-style post-training with SpaceMouse corrections, raised grasp-and-lift success from ~35% to 55%.

AWARDS AND GRANTS

Emory Wilson Family STEM Undergraduate Research Award *May 2026*
Awarded $1,200 to two outstanding undergraduate students to recognize and support their research activities.
Emory Pathway Global Internship Award *April 2025*
Awarded $2,500 for outstanding summer internship performance.
Mathematical Contest in Modeling (MCM), Honorable Mention *May 2025*
Emory Phi Sigma *May 2024*

LEADERSHIP, PROFESSIONAL DEVELOPMENT & OUTREACH

Emory University Mathematics Association (EUMMA) *President*
- Led 100+ club members and launched "AI4SE" to promote AI literacy through workshops and mentorship.
- Organized research and career panels; Led monthly training in AI and modeling for Kaggle and math competition.

Emory Student Government Association (SGA) *First Year Councilor*
- Elected to represent first-year class to gather feedback, voice student concern, and support campus life transition.

MIT AwesomeMath Summer Session, Algebra & Geometry *Teaching Assistant*
Atlanta Mathematics of Data Science Bootcamp *Student Volunteer & Attendee*
Atlanta Science Festival, Exploration Exposition *Student Volunteer*

SKILLS AND INTERESTS

Skills: Python, PyTorch, MATLAB, R, SQL, Git, Java, C, HTML. Toolkit: Docker, SSH, AWS
Research Interests: Agentic AI & RL, Foundation Model, robust learning systems for complex system, Scientific ML